

Complementos a dos

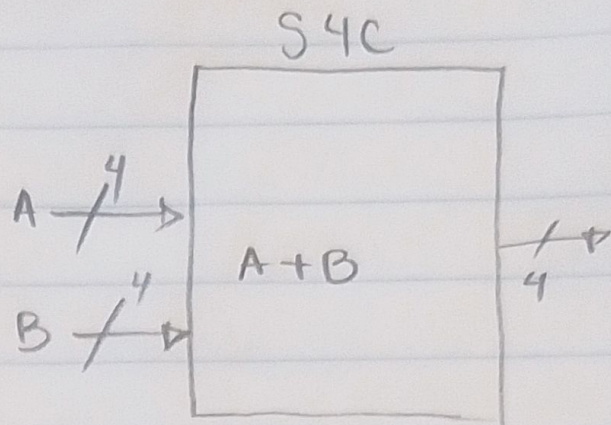
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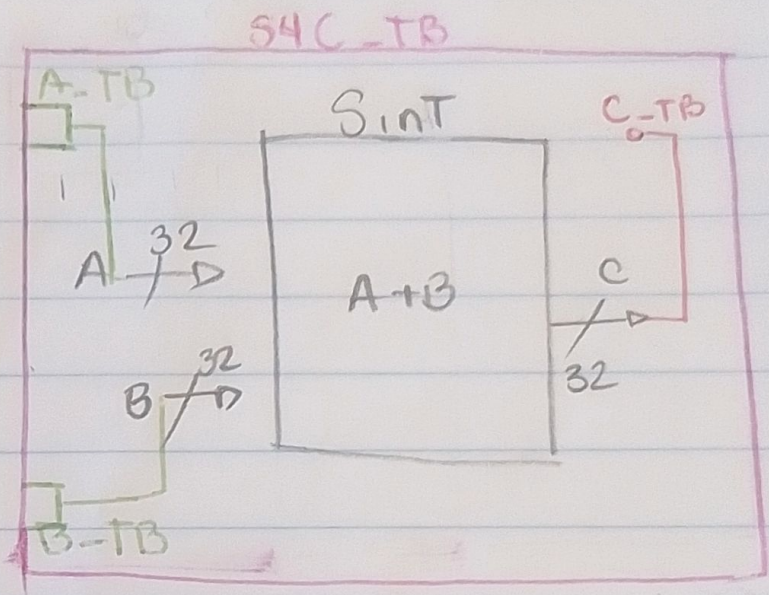
Scribe

comportamental

```
module S4C (
    input [3:0] A,
    input [3:0] B,
    output [3:0] C
);
    assign C = A + B;
end module
```



```
module Sint (
    input [31:0] A,
    input [31:0] B,
    output [31:0] C
);
    assign C = A + B;
end module
```



* Para números negativos

$$1001 = 9$$

$$1 \times 2^0 = 1$$

$$0 \times 2^1 = 0$$

$$0 \times 2^2 = 0$$

$$1 \times 2^3 = 8$$

Numero binario
a numero decimal

Bit de Signo

Comp. a 1

Comp. a 2^{12}

Se invierte y
se suma 1

$$\begin{array}{r} 00011 \\ 11100 \\ \hline 11101 \end{array}$$

signed

$$\begin{array}{r} 1001 \\ 0110 \\ \hline 0111 \end{array}$$

$$\begin{array}{r} 5 \text{ A } 0101 \text{ (5)} \\ -7 \text{ B } 1001 \text{ (9)} \\ \hline -2 \rightarrow 1110 \text{ (14)} \end{array}$$

$$-7 \rightarrow 1001 \rightarrow 0101 +$$

$$\begin{array}{r} 1 \rightarrow 0001 \\ -6 \rightarrow 1010 \end{array}$$

$$\begin{array}{r} 0101 + \\ 1 \\ \hline 0110 \end{array} \rightarrow -6$$